EXP 1. 8-Puzzle Problem

Aim

To solve the 8-puzzle problem using Breadth-First Search (BFS) and reach the goal arrangement.

Algorithm

1. Start with the initial board and the goal board.
2. Insert the initial board into a queue and mark it as visited.
3. Remove the first board from the queue. If it is the goal, stop.
4. Find the blank cell (0) and generate valid moves by swapping it with an adjacent cell.
5. Add each unvisited board to the queue and record its parent.
6. After finding the goal, trace the parent links and print the path from start to goal.

Program

AI.py - C:/Users/Barath/AppData/Local/Programs/Python/Python313/AI.py (3.13.15)

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from collections import deque

start = (1, 2, 3, 4, 0, 6, 7, 5, 8)
goal = (1, 2, 3, 4, 5, 6, 7, 8, 0)

q = deque([start])
parent = {start: None}

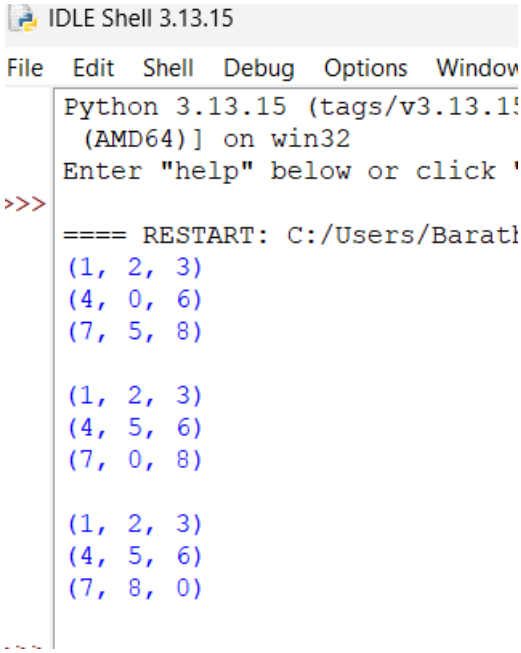
while q:
    s = q.popleft()
    if s == goal:
        break

    z = s.index(0)
    r, c = divmod(z, 3)

    for dr, dc in [(1, 0), (-1, 0), (0, 1), (0, -1)]:
        nr, nc = r + dr, c + dc
        if 0 <= nr < 3 and 0 <= nc < 3:
            nz = nr * 3 + nc
            a = list(s)
            a[z], a[nz] = a[nz], a[z]
            t = tuple(a)
            if t not in parent:
                parent[t] = s
                q.append(t)

if goal in parent:
    path = []
    s = goal
    while s is not None:
        path.append(s)
        s = parent[s]
    for s in reversed(path):
        print(s[:3])
        print(s[3:6])
        print(s[6:])
        print()
else:
    print("No solution")
```

Output



Result

Thus, the 8-puzzle problem was solved using BFS, and the goal state was reached.